

Heart Smart Tech: Navigating the Future with IoT-Enhanced Data Compression

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DECLARATION

We hereby declare that this project work titled “**Heart Smart Tech: Navigating the Future with IoT-Enhanced Data Compression**” is my original work and no part or it has been submitted for any other degree purpose or published in any other form till date.

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Signature of The Guide

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ABSTRACT

The perpetual rise in energy consumption represents a critical bottleneck in the seamless functioning of wireless sensor networks, a challenge exacerbated by the inherent constraints confronting these intricate systems. The diminutive stature of sensor nodes, characterized by limited memory storage, compact battery capacity, restricted processing capabilities, and bandwidth limitations, collectively contributes to the complexity of this predicament. This research is motivated by the imperative to address these challenges comprehensively, with a particular focus on pioneering data compression techniques as strategic interventions.

In response to the intricate web of challenges, data compression emerges as a pivotal strategy poised to curtail energy consumption and amplify the overall lifespan of wireless sensor networks. The crux lies in the strategic reduction of data size before transmission from the sensing node to the sink node. This nuanced application of data compression assumes instrumental significance as it effectively mitigates the resource constraints inherently associated with wireless sensor networks.

This research endeavors to conduct a thorough exploration into the multifaceted intricacies of energy consumption challenges within wireless sensor networks. A specialized emphasis is placed on the efficacy of cutting-edge data compression techniques as a viable and transformative solution, not only to alleviate the energy burden but also to fortify the sustainability and operational efficiency of wireless sensor networks in the broader context of the burgeoning Internet of Things (IoT) landscape.

CHAPTER 1

INTRODUCTION

1.1 Scope of Heart Disease

The fetal origins hypothesis states that fetal undernutrition in middle to late gestation, which leads to disproportionate fetal growth, programmes later coronary heart disease. Animal studies have shown that undernutrition before birth programmes persisting changes in a range of metabolic, physiological, and structural parameters [1]. Heart disease, a collective term encompassing a spectrum of cardiovascular conditions, emerges as a profound and pervasive global health concern. This umbrella category includes coronary artery disease, heart failure, valvular diseases, and various cardiac abnormalities. Its prevalence is staggering, contributing substantially to morbidity and mortality worldwide. Factors such as sedentary lifestyles, unhealthy dietary habits, and genetic predispositions contribute to the escalating incidence of heart disease. Heart disease is one of the complex diseases and globally many people suffered from this disease. On time and efficient identification of heart disease plays a key role in healthcare, particularly in the field of cardiology. [2]. Early detection and intervention strategies become paramount in mitigating the far-reaching consequences of heart disease. The multifaceted nature of its manifestations underscores the importance of comprehensive research, public health initiatives, and healthcare policies aimed at prevention, education, and improved treatment modalities.

1.2 Escalating Prevalence

Nowadays, heart disease is the leading cause of death worldwide. Predicting heart disease is a complex task since it requires experience along with advanced knowledge. Internet of Things (IoT) technology has lately been adopted in healthcare systems to collect sensor values for heart disease diagnosis and prediction. Many researchers have focused on the diagnosis of heart disease, yet the accuracy of the diagnosis results is low. To address this issue, an IoT

framework is proposed to evaluate heart disease more accurately using a Modified Deep Convolutional Neural Network (MDCNN). The smartwatch and heart monitor device that is attached to the patient monitors the blood pressure and electrocardiogram (ECG) [3]. The relentless surge in the prevalence of heart disease presents an increasingly urgent global health crisis, affecting individuals, healthcare systems, and societies at large. Lifestyle factors, including sedentary behaviors and unhealthy dietary patterns, contribute significantly to the escalating rates of cardiovascular conditions. Coupled with the rising incidence of obesity and an aging population, these factors create a perfect storm for the widespread occurrence of heart diseases such as coronary artery disease, heart failure, and various cardiac disorders. The implications of this surge are not confined to individual health alone; they reverberate across healthcare systems struggling to meet the growing demand for cardiovascular care. The strain on medical resources, both human and financial, poses challenges for delivering timely and effective interventions. Societal repercussions include diminished workforce productivity, as cardiovascular diseases often strike individuals in their prime working years, leading to increased absenteeism and disability. Moreover, the economic burden associated with the management of heart diseases places significant stress on national healthcare budgets. To address this burgeoning crisis, a comprehensive approach is imperative, encompassing public health campaigns to promote heart-healthy lifestyles, increased access to preventive care, and advancements in medical research and technology for more effective treatments.

1.3 Objective of the Report:

The primary objective of this report is to comprehensively explore the multifaceted aspects of heart disease through the following key points:

- **Prevalence:** Investigate and analyze the increasing rates of heart disease on a global scale.

- **Risk Factors:** Examine the diverse range of risk factors contributing to the development of heart diseases, including lifestyle choices, genetic predispositions, and environmental influences.
- **Symptoms:** Provide insights into the symptoms associated with various cardiovascular conditions, emphasizing the importance of early identification for effective intervention.
- **Diagnostic Methods:** Elaborate on the diagnostic methods employed in the detection and assessment of heart diseases, highlighting advancements in medical technology.
- **Treatment Options:** Explore the diverse range of treatment options available, spanning lifestyle modifications, medication, and advanced surgical interventions.

1.4 Significance of Data Collection:

- Beyond its immediate impact on individuals, the data collected holds profound significance for the broader scope of human society. Aggregated and anonymized data provides a comprehensive view of heart disease trends, prevalence, and contributing factors at a population level. This information becomes a crucial asset for public health planners, policymakers, and researchers. It aids in the formulation of evidence-based strategies for disease prevention and management, guiding the allocation of healthcare resources, and shaping educational campaigns. Sensors are used to detect the stench of people in the first place. There are three distinct kinds of glands in human skin, and it is these glands that are responsible for excreting volatile organic chemicals, as well as the volatile compounds themselves, which do not vary over time. During the production of glands, bacterial activity occurs on the human skin, causing an odor. Volatile organic compounds (VOCs) are excreted by people in large amounts, but only a small amount of these main volatile chemicals is required for this study, and the amount excreted has no relation to age or the surroundings [4].

CHAPTER 2:

LITERATURE REVIEW

Liang et al.[1] introduced the Convolution-based Transfer Learning CS (CTCS) model, employing transfer learning to reconstruct compressed signals. The evaluation, conducted using Ultrawide Band (UWB) radar echo signals and the Mnist handwritten dataset, demonstrated CTCS's superiority over traditional reconstruction algorithms in 6G-IoT. Robust performance was observed under varying noise levels, measurement numbers, and signal sparsities, with no discernible latency or reliability issues.

Zhang et al.[2] presented the AMP-Net, a deep unfolding model that avoids learning regularization terms by unfolding the iterative denoising process of the approximate message passing algorithm. Integrating deblocking modules to eliminate artifacts in compressed sensing (CS) of visual images, AMP-Net jointly trains the sampling matrix and network parameters, enhancing reconstruction performance. Notably, the model's performance was not evaluated under noisy scenarios.

Jiang et al.[3] proposed the Compressed Sensing with Dynamic Retransmission (CSDR) algorithm, prioritizing high data reconstruction accuracy, network lifetime, and energy utilization for IoT devices. The CSDR algorithm dynamically adjusts max packet loss retransmission times based on residual energies, optimizing network lifetime for energy-sensitive devices and improving data transmission accuracy for energy-rich ones. The study revealed the potential for utilizing remaining energy in distant sink areas to enhance network performance.

Zhang et al.[4] highlighted the impact of packet loss on CS algorithm performance in Wireless Sensor Networks (WSNs). Their approach involves dividing the WSN into clusters, with cluster heads responsible for collecting compressed sampling data. While limiting the impact of errors to one cluster, this method doesn't substantially reduce packet loss or analyze its broader implications on CS, affecting performance in scenarios with frequent packet loss.

Chen et al.[5] engineered a Two-Stage Dynamic Topology Compressed Sensing (DTCS) data collection algorithm for dynamic topology in wireless sensor networks. The first stage constructs a collection tree using the maximal independent set method, while the second stage involves nodes communicating for local topology information and adapting the collection tree accordingly.

Ramachandra et al.[6] explored the application of compressed sensing (CS) in image/video acquisition for IoT applications facing constraints in energy, bandwidth, and storage. The CS paradigm, incorporating simultaneous sensing and compression, emerged as an efficient technique for overcoming limitations.

Sun et al.[7] proposed an abnormal event detection scheme based on compressed sensing for Wireless Sensor Networks (WSNs). Considering low energy consumption targets, the sparse model of terminal nodes was introduced. When data collection exceeds the cluster head's capacity, a distributed abnormal event detection scheme utilizing compressed sensing and LDPC code-based rules was employed. The Message Passing algorithm based on Tanner Graph reconstructed abnormal event sensing results at the fusion center.

Compressed Sensing (CS) has a wide array of applications in signal processing, particularly in biomedical enhancement, compression, and recovery.[12] It has been specifically applied to ECG signals, assuming these signals are compressible. CS is considered highly efficient in terms of energy usage due to its reduced complexity and

lower CPU execution time. The compression phase in CS is quick, straightforward, and energy-efficient, making it preferred for various sensing applications. However, the recovery phase is complex and non-linear, requiring processors with significant memory, speed, and computational capability.

CS has emerged as an effective technique for reconstructing sparse signals, including ECG, by leveraging their inherent sparsity. Despite its advantages, CS faces certain limitations, which are being addressed through the integration of deep learning methods:

- **Reconstruction Methods and Algorithms:** Traditional CS reconstruction techniques like Orthogonal Matching Pursuit (OMP) and Basis Pursuit (BP) have computational constraints, especially for high-dimensional ECG signals. Recent research has demonstrated that deep learning techniques,
- such as Recurrent Neural Networks (RNNs) and Convolutional Neural Networks (CNNs), can serve as accurate and efficient reconstruction algorithms, reducing computational complexity and improving reconstruction quality.
- **Robustness to Noise:** ECG signals often contain various noise and artifacts that degrade the quality of CS. Deep learning methods can automatically learn robust features and represent noisy data more effectively, enhancing the reliability of ECG reconstruction.
- **Adaptability:** CS typically uses a fixed sensing matrix and assumes a particular sparse basis for the signals. However, the sparse structure of ECG signals can vary between individuals and over time. Deep learning models can learn data-driven representations and sparsity structures, improving reconstruction performance by adapting to these variations.
- **Combining Multiple Data Sources:** CS aims for efficient signal acquisition and reconstruction. By combining ECG signals with other physiological data (such as respiration, blood pressure, and motion data), deep learning can provide a more comprehensive analysis of a patient's medical condition, thereby enhancing performance.

Researchers are integrating deep learning methodologies with CS to overcome these limitations, aiming to improve accuracy, efficiency, and robustness in ECG signal processing for healthcare IoT systems. This approach can significantly enhance patient monitoring and care. Recent frameworks, such as Consistent Label Joint Crossed Domain Data Learning (LC-XDJDL) and Joint Crossed Domain Data Learning (XDJDL), have been developed to improve ECG quality and diagnosis based on photoplethysmography (PPG) [13][14]. These frameworks optimize ECG and PPG signals by leveraging their sparse codes and disease-related information.

Innovative wearable IoT devices for exercise, rehabilitation, and monitoring have also been introduced. These systems evaluate training efficiency and record physical parameters like ECG, electromyogram (EMG) signals, and body temperature. Van-Sloun [15] et al. developed a multimodal data combination technique to enhance beat detection accuracy in ambulatory monitoring. This technique isolates PPG and ECG signals using wavelet discrete transform (DWT) and generates a combined signal feature for maximum detection.

Another example is the ultra-low-energy ECG monitor developed by Ianni and Airan [16], designed as a lightweight, wrist-worn IoT device that tracks ECG waveforms, heart rate, location, and physical activity. The SPHERE platform evaluates patients' health conditions by collecting data in residential environments. Economical Internet of Medical Things (IoMT) ECG devices have been developed for real-world applications to record and monitor heart abnormalities. To reduce noise in non-medical equipment signals, a cloud-based denoising technique using deep neural networks and the frequency-time domain is employed. The ECG signal is transformed using the Stockwell Fractional Transform (FrST) and processed through a robust deep dual-step network (Deep RSTN) to eliminate noise.

Furthermore, task-specific sparsity-aware signal processing methods have been utilized to retrieve essential information. For instance, length-adaptive correlation-

assisted compression (CCAL) separately compresses ECG signals for each pseudo-period, preserving individual heartbeat information and dynamically adjusting compression length by evaluating coefficients across adjacent sections. Despite advancements, challenges remain in ECG signal processing, such as noise robustness, high-dimensional data scalability, and integration with various physiological data. Addressing these issues can significantly enhance ECG recording and monitoring accuracy and efficiency.

The growing demand for effective ECG signal processing using IoT systems has driven the exploration of various approaches. Combining deep learning methods with autoencoders and CS has shown promise in achieving accurate and efficient ECG signal reconstruction. CS reduces data acquisition and transmission needs by exploiting ECG signal sparsity, while autoencoders accurately learn the required reconstruction representations. Deep learning methodologies further enhance computational efficiency and accuracy in the reconstruction process. These combined methods address limitations in healthcare IoT systems, leading to notable improvements in patient monitoring and care.

Energy efficiency in data collection is a primary concern in wireless sensor networks (WSNs). This issue has been addressed through various strategies, including energy-conserving sleep scheduling, topology control, mobile data collectors, and data aggregation. While the first three strategies focus on improving the efficiency of data transport within the network, data aggregation specifically aims to reduce the volume of data that needs to be transmitted. This makes it a crucial mechanism for achieving energy-efficient data collection in WSNs.

Despite extensive research into data aggregation techniques, there are still areas that need improvement. Firstly, lossy aggregation methods that use simple functions like MIN, MAX, and SUM only capture certain statistical aspects of the sensory data, leading to loss of other valuable information. Consequently, these techniques are only

suitable for applications requiring limited information. Secondly, although distributed source coding techniques like Slepian-Wolf coding theoretically allow for non-collaborative data compression at the source, their practical application is limited by the need for prior knowledge of data correlation structures. Lastly, while collaborative in-network compression can reveal data correlation structures through information exchange, it either requires a simple correlation structure or results in high communication loads that can negate its benefits.

Compressed sensing (CS) is a relatively new signal processing technique that promises to fully recover signals with high probability from far fewer measurements than their original dimension, provided the signals are sparse or compressible in some domain. Although CS suggests a way to reduce sensory data traffic in WSNs without needing to adapt to the data correlation structure, the complexity of integrating data routing with CS-based aggregation has delayed progress in this area until recently. This paper [17] introduces a new data aggregation technique based on CS, aimed at minimizing the total energy consumption in collecting sensory data across the entire network. To our knowledge, this is the first study to explore the minimum energy CS-based data aggregation problem within a combinatorial framework.

A mixed-integer programming (MIP) [17] formulation of MECDA is presented, which can obtain optimal solutions for small-scale WSNs. Additionally, a greedy heuristic is proposed that delivers near-optimal solutions for large-scale WSNs. Extensive numerical results validate the efficacy of this heuristic and demonstrate the energy efficiency of the CS-based data aggregation approach.

However, several fundamental technical challenges remain in applying CS to Body Sensor Network (BSN) systems [18]. One major concern is that analog signals in modern computing systems must be digitized before wireless transmission, and the impact of quantization in CS on the energy-performance tradeoff is still under-explored. Different components of a BSN system have distinct design priorities: distributed body sensors are typically constrained by energy limitations and are highly

sensitive to energy consumption, whereas data aggregators focus on compiling high-quality data.

Compressed Sensing (CS) has also been used in scenarios where recharging or replacing batteries is difficult or impossible [19], such as in wireless sensor networks (WSNs) for water distribution systems. Energy efficiency is crucial in these networks due to the impracticality of battery maintenance. CS provides an efficient method to reconstruct signals from limited samples, making it a valuable technique for extending network lifetimes through optimized sensor activation schemes.

To mitigate the risks of water pollution and pipeline leakages [19][20], WSNs have been deployed in various cities to monitor water distribution networks. These networks are a typical example of the Internet of Things (IoT). Since pipelines are located underground, replacing sensor node batteries is both inconvenient and costly. Additionally, WSNs must ensure good sensing performance, such as timely measurements and accurate estimations, introducing a natural tradeoff between energy efficiency and monitoring performance.

The miniaturization and reduced cost of water sensors have facilitated the deployment of dense, spatially distributed sensor networks for collecting hydraulic and water quality data. Dense WSNs leverage redundant sensor nodes to minimize the duty cycle of each individual sensor, thereby prolonging network lifetime and enhancing robustness against sensor failures. By scheduling sleep periods for sensor nodes, energy efficiency can be improved. Under this approach, sensor nodes can be uniformly deployed over the target area, and an activation schedule can be determined to ensure that the network remains connected and maintains satisfactory monitoring performance.

Given the high correlation of measurements in dense sensor networks, network lifetime can be significantly extended with minimal loss in monitoring accuracy.

Building on prior work, a CS-based activation scheme has been devised to reduce the number of active sensor nodes, thereby decreasing energy consumption and balancing the residual energy among sensor nodes. Energy balancing helps achieve maximum network lifetime and robustness to sensor node failures, which is particularly important in underground pipeline systems where battery replacement is challenging.

In a densely deployed WSN, only a subset of sensor nodes is scheduled to sense and transmit data to sink nodes in a multi-hop fashion at any given time. The goals are to balance the energy consumption of sensor nodes, ensure accurate monitoring and fast response, and maintain connectivity among activated sensor nodes so that sensed data can reliably reach the sink nodes.

In-network data compression in wireless sensor networks (WSNs) leverages the spatial correlation of sensor readings to reduce data transmission. Conventional compression techniques, such as joint entropy coding, use relayed data as side information to encode sensor readings more efficiently. This method reduces the number of bits needed for encoding because correlated readings allow for more efficient compression. However, it only uses unidirectional data correlation. Another approach, cooperative transform compression, involves nodes performing transformations like distributed wavelet transform. Even nodes broadcast their readings, and odd nodes compute high-pass coefficients and send them back to even nodes, which then compute low-pass coefficients. This method, while effective in utilizing correlation, incurs significant communication overhead. Quantization-based compression techniques, like Clustered Aggregation (CAG) [22], form clusters based on sensing values and transmit one representative reading per cluster, thereby reducing data transmission while maintaining an acceptable error threshold.

However, conventional compression techniques face significant challenges. Their performance heavily relies on how routes are organized, necessitating joint optimization of compression and routing algorithms, which is computationally

complex. Additionally, the efficiency of these techniques is affected by the computational and communication overheads involved.

Distributed source coding techniques aim to reduce complexity at sensor nodes by utilizing the correlation at the sink. Based on the Slepian-Wolf [23] coding theory, these techniques independently compress correlated readings but decode them jointly, eliminating the need for data exchanges and decoupling routing from compression. Despite their advantages, these techniques face challenges in large-scale WSNs because they require knowledge of the global correlation structure to allocate the appropriate number of bits. Moreover, distributed source coding performs well for static correlation patterns but struggles when correlation patterns change or during abnormal events, leading to significant decoding errors.

Compressive sampling theory offers a novel approach to in-network data compression, promising benefits like distributed source coding, such as simple encoding and reduced inter-node data exchange. It has the added advantages of gracefully handling abnormal sensor readings and being resilient to packet losses, as all messages received by the sink are equally important. Compressive Wireless Sensing (CWS) reduces data gathering latency by delivering linear projections of sensor readings through synchronized transmissions, although synchronization challenges limit its practicality for large-scale networks. Nonetheless, compressive sampling holds potential for data aggregation in multi-hop WSNs, decoupling compression from routing and efficiently managing abnormal readings, making it a promising solution for large-scale WSN data gathering.

CHAPTER 3

COMPRESSIVE SENSING FUNDAMENTALS AND ITS DETAILS

Compressed Sensing (CS) is an innovative and increasingly important technique in signal processing, designed for efficiently acquiring and reconstructing signals that are either inherently sparse or can be made sparse through an appropriate transformation. The term "sparse" refers to signals that contain many zero or near-zero coefficients when represented in a certain basis. This property of sparsity is leveraged in CS to sample signals at a much lower rate than traditionally required by the Nyquist-Shannon sampling theorem, which states that a signal must be sampled at twice its highest frequency to be accurately reconstructed.

The field of compressed sensing has gained significant traction due to its potential to revolutionize various applications by reducing the number of measurements needed for accurate signal reconstruction. This reduction in the number of samples translates to faster acquisition times, lower storage requirements, and reduced computational burdens. These benefits make CS particularly attractive for applications where data acquisition is expensive, time-consuming, or resource intensive.

We assume x is an N -dimensional vector and sampled using M -measurement vector y

$$y = \Phi x \tag{1}$$

where $\Phi \in R^{M \times N}$ is the sensing array, which models the linear encoding, and M is defined as the sampling rate in N -dimensional CS. The elements in Φ are either

Gaussian random variables or Bernoulli random variables. Because of $M \ll N$, the formulation in (1) is undetermined, and signal x cannot be uniquely retrieved from sensing array Φ and measurements y . However, under a certain sparsity-inducing basis $\Psi \in R^{N \times N}$, the signal x can be represented by a set of sparse coefficients

$$\begin{aligned} u &\in R^N \\ x &= \Psi u \end{aligned} \quad (2)$$

i.e., the coefficient u , under the transformation Ψ , only has few non-zero elements. Therefore, based on (1) and (2), the sparse vector u can be represented as follows:

$$y = \Phi \Psi u = \Theta_{M \times N} u \quad (3)$$

where $\Theta_{M \times N} = \Phi \Psi$ is an $M \times N$ array, called measuring matrix. Due to the prior knowledge that the unknown vector u is sparse, it is possible to estimate the value u using the l_0 minimization formulation as follows:

$$u = \min \|u\|_0, \quad s.t. \|y - \Theta u\| < \varepsilon \quad (4)$$

where ε is the reconstruction error tolerance. The formulation in (4) is a determined system with unique solutions. However, l_0 is an NP-hard problem, and one of the methods to solve (4) is to approximate l_0 minimization formulation to l_1 minimization formulation

$$u = \min \|u\|_1, \quad s.t. \|y - \Theta u\| < \varepsilon \quad (5)$$

Under the condition of restricted isometry property (RIP) [25] l_1 has been theoretically proven to be equivalent to minimize l_0 . Moreover, l_1 minimization is convex and can be solved within polynomial time. In this work, we will use the l_1 -based approach CS. After estimating the sparse coefficient u with the formulation in (5), the original input signal x can be recovered directly

$$\hat{x} = \psi u \quad (6)$$

Various sophisticated data compression techniques have been meticulously explored and implemented to address the unique challenges posed by IoT (Internet of Things) devices. These techniques play a pivotal role in ensuring the efficient management

and transmission of data in environments characterized by resource constraints. Let's delve into the intricacies of some commonly employed compression methods:

3.1 Run-Length Encoding (RLE):

- The Run-Length Encoding (RLE) method presents an elegant solution within the realm of data compression by employing a mechanism that represents consecutive data values through a single code paired with a count parameter. Its efficacy is particularly conspicuous when confronting datasets characterized by protracted sequences of repeated values. Through the condensation of these sequences, RLE makes substantial contributions towards reducing data volume and enhancing transmission efficiency.

3.2 Huffman Coding:

- Employing a variable-length encoding methodology, Huffman Coding distinguishes itself by allocating shorter codes to data values that occur with greater frequency. This adaptive strategy guarantees a decrease in the aggregate size of data, rendering it particularly valuable for Internet of Things (IoT) applications where efficient resource utilization is critical. The elegance of Huffman Coding lies in its dynamic adaptability to the frequency distribution of the data it processes. This flexibility enables the encoding scheme to swiftly adjust to varying patterns within the data, thereby optimizing compression performance.

3.3 Delta Encoding:

- Designed with precision for scenarios marked by incremental shifts in data, Delta Encoding executes its function by capturing and storing the differences between consecutive data values rather than transmitting the actual values themselves. This innovative strategy proves particularly advantageous in bandwidth conservation during the transmission of data characterized by gradual, sequential changes over time. By focusing solely on the alterations between successive data points, Delta

Encoding significantly minimizes the volume of data transmitted, facilitating more streamlined and efficient communication processes.

4.4 Quantization:

- Operating at the granular level of numerical precision, Quantization strategically diminishes the precision of numerical data by rounding values to a specified level. Through this process, the technique substantially diminishes the number of bits necessary to represent the data accurately. Particularly in resource-limited Internet of Things (IoT) environments, where bandwidth is frequently scarce and expensive, Quantization stands out as an invaluable asset for reducing the size of data payloads. By judiciously sacrificing some precision in exchange for reduced data size, Quantization enables more efficient utilization of available resources and smoother data transmission within constrained network environments.

3.5 Dictionary-based Compression

- Harnessing the efficacy of lexicons, this compression methodology operates via the substitution of frequently recurring subsequences within the dataset with abbreviated codewords. This systematic process effectively mitigates redundancy within the data structure, resulting in a notable reduction in overall data size while concurrently optimizing the efficient allocation of available resources. Such resource optimization holds particular significance within the domain of Internet of Things (IoT) applications, where meticulous management of memory resources is imperative. By intelligently mapping repetitive patterns to concise representations, dictionary-based compression not only conserves precious memory space but also streamlines data transmission protocols, thereby ensuring the seamless functionality of IoT devices within resource-constrained operational environments.

3.6 Entropy Coding:

- Rooted in information theory, Entropy Coding employs a probabilistic approach to assign shorter codes to more probable data, leveraging statistical properties for compression. In the context of the Internet of Things (IoT), where data often exhibits patterns and predictability, Entropy Coding emerges as an intelligent choice for optimizing data representation. By delving into data distributions' probabilistic nuances, this method strategically crafts encoding schemes prioritizing compression of prevalent data elements while preserving fidelity. Analyzing the data's entropy enables tailored codeword assignments, optimizing compression ratios without compromising essential information. This approach enhances resource optimization and bandwidth conservation within IoT ecosystems, facilitating efficient data transmission and storage while maintaining theoretical rigor, making it a cornerstone technique for IoT data management.

3.7 Lempel-Ziv-Welch (LZW):

- The Lempel-Ziv-Welch (LZW) algorithm, a prominent dictionary-based compression technique, enjoys ubiquitous utilization, particularly in domains involving image processing and Unix file compression. Functioning on the principle of substituting commonly recurring patterns with abbreviated codes, LZW systematically diminishes the aggregate data size with remarkable efficiency. Its methodological essence lies in dynamically updating a dictionary as it encounters novel patterns within the data, subsequently replacing them with concise representations. This adaptability ensures that LZW optimally handles diverse datasets encountered in Internet of Things (IoT) applications, where data may exhibit intricate structures and varying degrees of redundancy. The algorithm's prowess in efficiently encoding data, coupled with its versatility across different data types and formats, solidifies its standing as a foundational component in numerous IoT scenarios. LZW's enduring relevance underscores its capacity to address the evolving demands of IoT ecosystems, where resource optimization and data compression are pivotal for seamless operation and efficient resource utilization.

In the dynamic ecosystem of the Internet of Things (IoT), where devices function within resource-constrained environments, the selection of an appropriate compression technique emerges as a pivotal decision. This choice is influenced by a multitude of factors, encompassing the intrinsic characteristics of the data, the specific exigencies of the IoT application, and the inherent limitations of the device itself, encompassing considerations such as processing power, memory capacity, and energy consumption. Each compression method, whether rooted in dictionary-based approaches, entropy coding, or numerical precision reduction, presents its own set of advantages and limitations, contributing distinctively to the overarching objective of optimizing data efficiency within IoT ecosystems.

The efficacy of a compression technique hinges on its ability to harmonize with the unique demands of the IoT landscape. For instance, dictionary-based methods excel in scenarios where data exhibits repetitive patterns, such as image processing or Unix file compression. Conversely, entropy coding techniques, drawing upon probabilistic principles, offer significant gains when dealing with data characterized by varying probabilities of occurrence. Additionally, strategies like delta encoding and run-length encoding prove invaluable for capturing incremental changes in data or condensing prolonged sequences of repeated values, respectively.

Moreover, the selection of a compression technique necessitates a judicious balance between compression efficiency and computational overhead. While some methods may achieve remarkable compression ratios, they may also entail substantial computational requirements, posing challenges for resource-constrained IoT devices. Hence, the suitability of a compression approach is contingent upon striking a delicate equilibrium between compression efficacy, computational complexity, and resource utilization.

In the realm of Internet of Things (IoT), the selection of an appropriate compression technique constitutes a nuanced decision-making process, necessitating a thorough examination of numerous intricate factors. Foremost among these considerations is an

in-depth analysis of the inherent characteristics of the data to be compressed. This entails scrutinizing factors such as data volume, variability, redundancy, and predictability, all of which significantly influence the choice of compression methodology. Furthermore, the specific requirements and objectives of the intended IoT application play a pivotal role in shaping the compression strategy. Whether the application demands real-time data transmission, extensive data storage, or energy-efficient operation, these considerations inform the selection of a compression technique tailored to meet the precise needs of the application.

Equally crucial are the constraints imposed by the IoT devices themselves, encompassing parameters such as processing power, memory capacity, energy consumption, and bandwidth limitations. The chosen compression approach must strike a delicate balance between achieving optimal compression ratios and minimizing computational overhead to ensure compatibility with the capabilities of the targeted IoT devices. Additionally, considerations extend to the operational environment, encompassing factors such as network reliability, latency, and security requirements, all of which influence the choice of compression technique.

By navigating these multifaceted considerations with precision and foresight, stakeholders within the IoT domain can identify and implement compression strategies that not only optimize data efficiency but also foster seamless communication and enhance the overall performance of IoT deployments. Through meticulous assessment and strategic decision-making, stakeholders can harness the full potential of compression technologies to mitigate bandwidth constraints, conserve energy, streamline data transmission, and ultimately unlock new frontiers of innovation within the burgeoning landscape of IoT applications.

CHAPTER 4

DATA ACQUISITION

In the intricate landscape of healthcare analytics, Data Acquisition emerges as a pivotal process, systematically capturing a diverse spectrum of vital health parameters. This comprehensive dataset encompasses not only the physiological metrics like pulse, blood pressure, and cholesterol levels but also delves into lifestyle intricacies such as smoking habits, physical activity, dietary patterns, alcohol consumption, and the presence of sleep apnea. This exhaustive approach provides a holistic view of an individual's health, recognizing that health is a complex interplay of various factors.

Augmenting this, demographic particulars including age and gender contribute to the multifaceted understanding of an individual's health profile. These demographic elements add contextual layers to the data, acknowledging that health outcomes can be influenced by age-related factors and gender-specific nuances.

The dataset extends its reach to encompass advanced medical diagnostics like ECG and SPO2, delving deeper into the physiological aspects that can influence heart health. The inclusion of ECG (Electrocardiogram) data allows for a detailed analysis of the heart's electrical activity, providing insights into cardiac health. Simultaneously, SPO2 (Oxygen Saturation) data sheds light on the oxygen levels in the blood, contributing valuable information about respiratory and circulatory functions.

The critical factor of the presence or absence of diagnosed heart disease adds a diagnostic layer to the dataset. This binary distinction forms a cornerstone in identifying individuals with pre-existing heart conditions, enabling targeted interventions and personalized healthcare approaches. The orchestration of this automated data assembly is facilitated by specialized sensors, a symphony of technology featuring MAX30100/102, DSI81320, DHT11, AD8232, and HC SR04. These sensors ensure precision and efficiency in data collection, contributing to the reliability and accuracy of the dataset.

Each parameter within this ensemble assumes significance, as its nuances unfurl in the tapestry of heart health, enabling a meticulous analysis. The intricate interplay of these diverse parameters provides a rich source of information, laying the groundwork for a nuanced understanding of an individual's cardiovascular well-being.

This robust foundation of data acquisition stands as the bedrock for the integration of sophisticated data compression techniques, meticulously tailored to elevate the analysis and diagnosis of heart-related conditions within the dynamic realm of the Internet of Things (IoT) in healthcare. The fusion of advanced data acquisition with cutting-edge compression methodologies promises to usher in a new era of personalized and data-driven healthcare, where insights gleaned from diverse parameters pave the way for targeted interventions and improved patient outcomes.

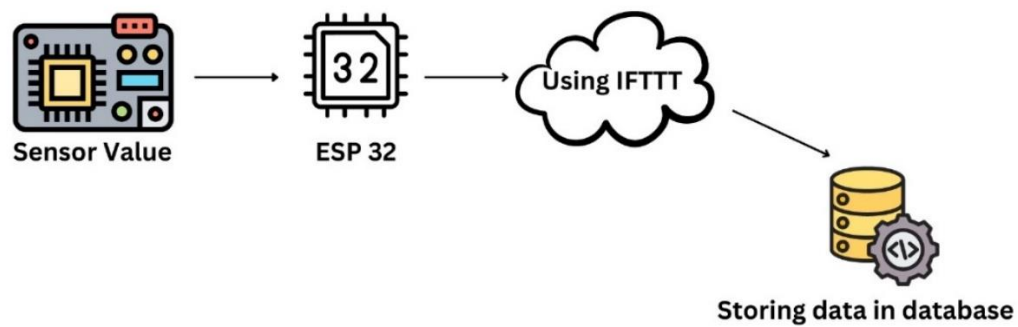


Figure 1: Data Sending Path from different Sensors to Backend using IFTTT

Pulse:

- Description: Pulse represents the heart rate, indicating the number of heart beats per minute.
- Impact on Heart Disease: Abnormal pulse rates may suggest irregularities in heart function or potential cardiovascular issues.

Blood Pressure:

- Description: Blood pressure measures the force exerted by circulating blood on the walls of blood vessels.
- Impact on Heart Disease: High blood pressure can strain the heart and arteries, increasing the risk of heart disease.

Cholesterol:

- Description: Cholesterol levels in the blood, specifically LDL and HDL, are key indicators.

- Impact on Heart Disease: Elevated LDL (bad cholesterol) levels can contribute to plaque buildup in arteries, leading to heart issues.

Smoking:

- Description: Indicates the habit of smoking.
- Impact on Heart Disease: Smoking is a major risk factor for heart disease, constricting blood vessels and reducing oxygen supply to the heart.

Fasting Blood Sugar:

- Description: Measures glucose levels in the blood after fasting.
- Impact on Heart Disease: Elevated blood sugar levels may contribute to the development of heart disease over time.

Height (cm) and Weight:

- Description: Physical measurements to assess body mass.
- Impact on Heart Disease: Body mass index (BMI) is derived from these values, and obesity is linked to heart disease.

Obesity and BMI:

- Description: Reflects the presence of obesity and the Body Mass Index.
- Impact on Heart Disease: Obesity is associated with increased heart disease risk due to higher strain on the cardiovascular system.

Physical Activity:

- Description: Quantifies the level of physical exercise.

- Impact on Heart Disease: Regular physical activity is a protective factor, reducing the risk of heart disease.

Diet:

- Description: Encompasses nutritional habits.
- Impact on Heart Disease: Unhealthy diets high in saturated fats and cholesterol contribute to heart disease.

Alcohol Consumption:

- Description: Measures the frequency and quantity of alcohol intake.
- Impact on Heart Disease: Excessive alcohol consumption can elevate blood pressure and contribute to heart issues.

Sleep Apnea:

- Description: Indicates interruptions in breathing during sleep.
- Impact on Heart Disease: Sleep apnea is linked to hypertension and an increased risk of heart disease.

Age and Gender:

- Description: Demographic information.
- Impact on Heart Disease: Age and gender influence heart disease risk, with older age and male gender being associated with higher risk.

ECG and SPO2:

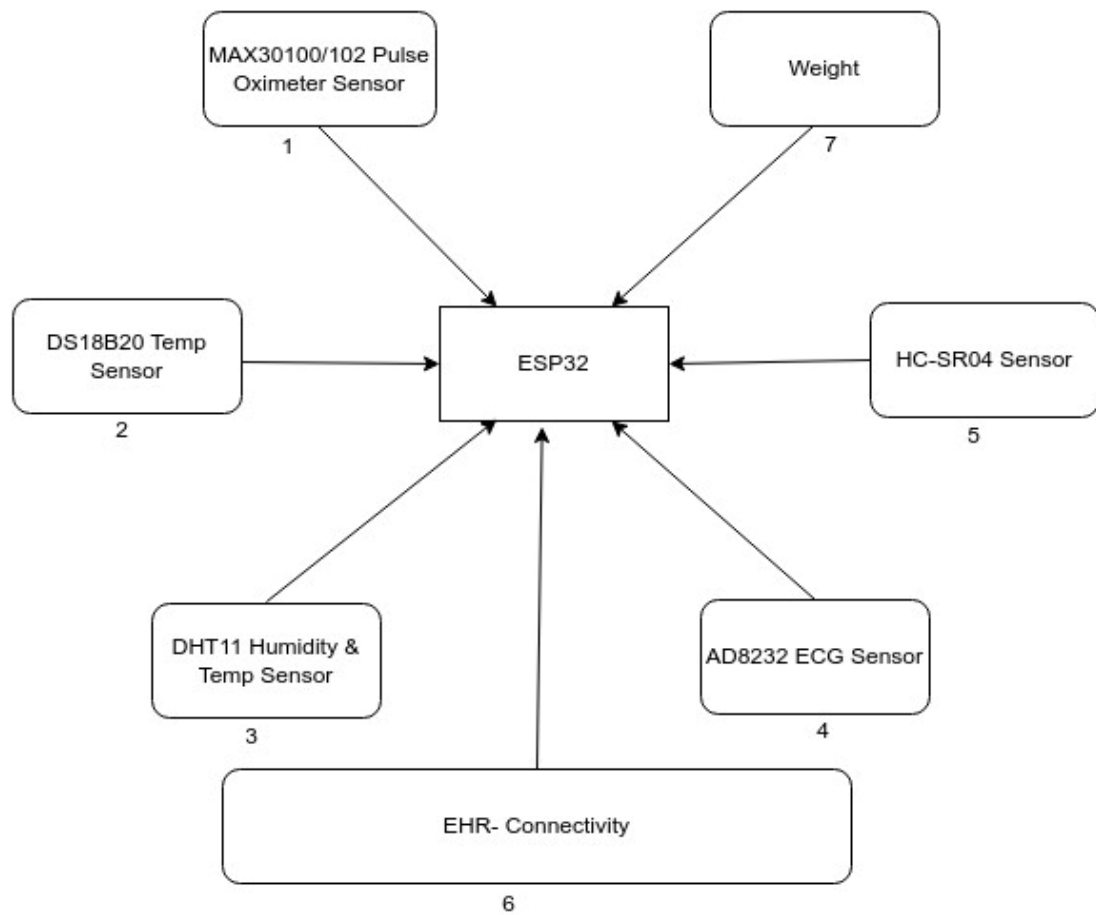
- Description: ECG monitors heart's electrical activity; SPO2 measures blood oxygen levels.

- Impact on Heart Disease: Irregular ECG patterns or low SPO2 levels may indicate cardiac issues.

Heart Disease:

- Description: Presence or absence of diagnosed heart disease.

- Impact on Heart Disease: A crucial variable representing the primary outcome, aiding in assessing the effectiveness of data compression techniques.



Sensor	Work
1. MAX30100/102 Pulse Oximeter	Read Oxygen Level and Pulse Rate Measurement
2. DS18B20 Temp Sensor	Read body Temp
3. DHT11 Humidity & Temp Sensor	Read Room Temp and Humidity Sensor
4. AD8232 ECG Sensor	For the electrocardiography
5. HC-SR04 Sensor	For Height Measurement
6. EHR- Connectivity	For data sharing

Figure 2: Sensors Connectivity.

The data acquisition process is facilitated by specialized sensors like MAX30100/102, DSI81320, DHT11, AD8232, and HC SR04, ensuring automated and accurate collection of diverse health metrics. This comprehensive dataset forms the foundation for developing advanced data compression strategies tailored for heart disease analysis in the era of IoT-enabled healthcare.

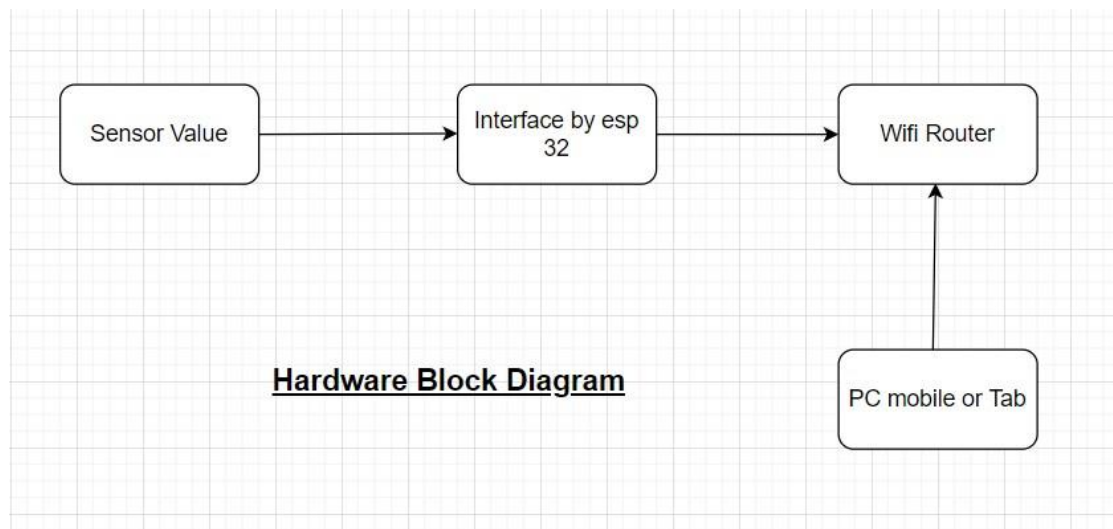


Figure 3: Hardware Block Diagram

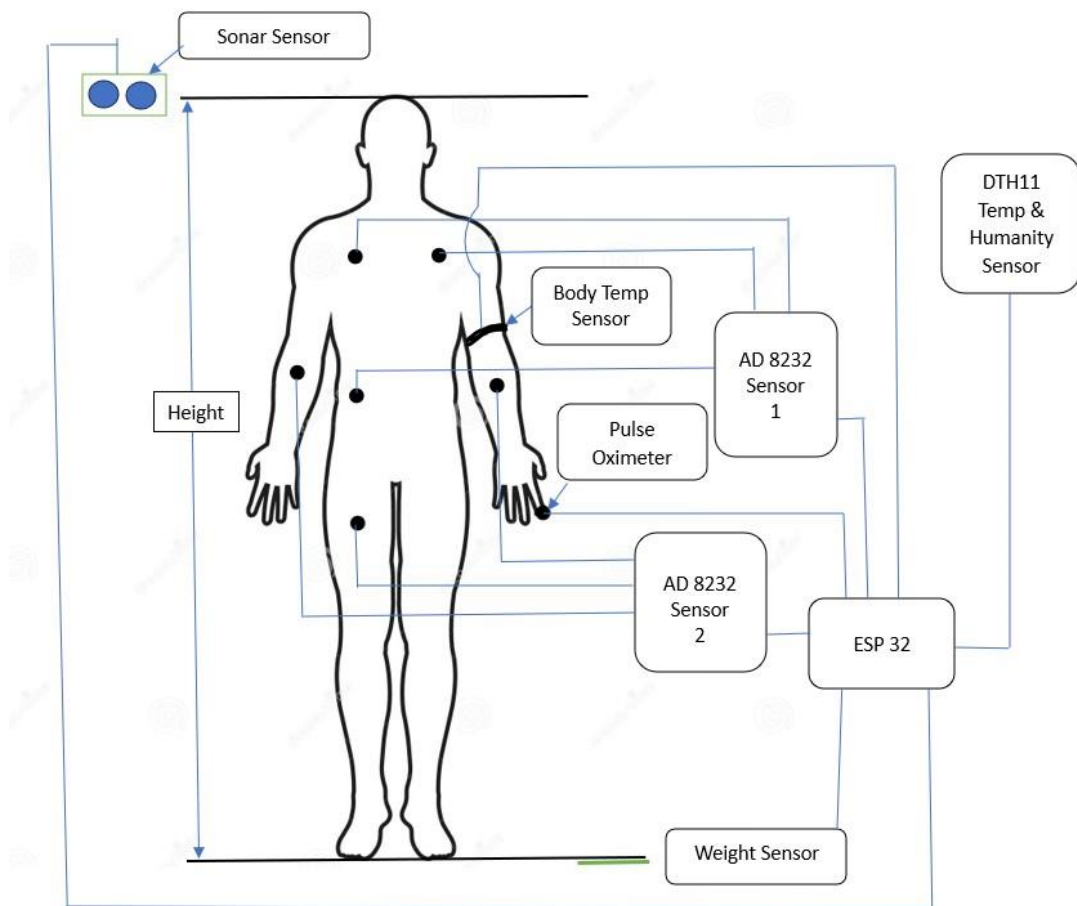


Figure 4: Interconnection of Sensors Within the Human Body.

In our sensor network architecture, each sensor is equipped with an ESP32 microcontroller, designated as the sensor node, which continuously monitors environmental parameters. These sensor nodes transmit their readings to a central ESP32 microcontroller, designated as the coordinator, responsible for aggregating and processing the collected data. The communication between the sensor nodes and the coordinator is facilitated through GSM technology, ensuring reliable data transmission over potentially vast distances. Each sensor node transmits minimal data, specifically a 1-bit signal indicating the status of the sensor reading. This minimalistic data transmission approach is essential for conserving energy and bandwidth, critical for maintaining the longevity and reliability of the wireless sensor network (WSN).

The data transmission protocol initiates with each sensor node persistently monitoring its designated environmental parameter, such as temperature, humidity, or pressure. If the sensor reading remains within a predefined acceptable threshold range, the sensor node transmits a 1-bit value of 0 to the coordinator. This bit value signifies that there has been no significant deviation in the monitored parameter. Conversely, if the sensor reading deviates outside the acceptable range, the sensor node transmits a 1-bit value of 1, indicating an anomaly or a significant change requiring further attention.

Upon receiving data from the sensor nodes, the coordinator employs multiplexing to efficiently manage the concurrent data streams. Multiplexing allows the coordinator to handle and process data from multiple sensor nodes by rapidly switching between the data streams, ensuring seamless integration and minimizing data latency. This technique is critical for the efficient operation of the WSN, especially in scenarios involving a large number of sensor nodes.

When a sensor node transmits a 1-bit value of 1, indicating an out-of-range reading, the transmission includes additional data to provide detailed context about the anomaly. If the sensed value exceeds the upper limit of the acceptable range, the sensor node transmits an additional 1-bit value of 1, followed by the quantitative difference between the sensed value and the upper limit. This two-step transmission

process ensures that the coordinator receives not only the indication of an anomaly but also the magnitude of the deviation. Similarly, if the sensed value falls below the lower limit, the sensor node transmits an additional 1-bit value of 0, followed by the difference between the lower limit and the sensed value, thus quantifying the degree of the deviation below the acceptable range.

The coordinator processes the incoming data to ascertain the current status of the monitored environment. For a received 1-bit value of 0, the coordinator registers no significant change and proceeds with routine operations. For a received 1-bit value of 1, the coordinator processes the subsequent data to evaluate the extent of the deviation, thereby enabling informed decision-making. This could involve triggering alerts, logging data for further analysis, or initiating corrective actions.

This structured and efficient communication protocol ensures accurate and timely environmental monitoring while optimizing energy consumption and bandwidth utilization. This approach is particularly advantageous for applications requiring continuous, long-term monitoring, such as environmental surveillance, industrial automation, and precision agriculture. The integration of GSM technology and multiplexing further enhances the network's capacity to manage numerous sensor nodes, rendering the system scalable, robust, and well-suited for extensive deployment scenarios.

CHAPTER 5

PROPOSED DESIGN

The system begins by initializing an array designed to store the boolean data received from various microcontrollers connected to the sensors. This array, with a size defined by `noOfSensors`, holds boolean values where each element can either be 0 or 1. Here, 0 signifies that the sensor reading is within the predetermined acceptable range, while 1 indicates that the reading is outside this range, signaling a potential issue that needs immediate attention.

Upon receiving data from any sensor node, if the initial flag input is set to 1, indicating an out-of-range reading, the system initiates a detailed monitoring process. This involves taking additional readings from the sensor over the next five instances to verify the consistency and persistence of the anomaly. During these five monitoring instances, the system continuously updates the array. If, at any point during these five instances, all elements in the array revert to 0, it indicates that the sensor readings have returned to the acceptable range, and no further action is required. This step is crucial as it helps in filtering out transient anomalies that might not require immediate intervention.

However, if any of the elements in the array remain at 1 by the end of the five instances, the system recognizes these as persistent out-of-bound readings. The system then counts the number of indexes in the array that are still set to 1, a count referred to as "outOfBound." This count provides a measure of how frequently the sensor readings have fallen outside the acceptable range during the monitoring period, offering insights into the severity and consistency of the anomaly.

If, after the five-instance monitoring period, any sensor readings remain out of the acceptable range, the system escalates the issue. The collected data, along with an SOS text, is sent to the local data center. This immediate communication ensures that

the anomaly is promptly flagged for further investigation and action by the local authorities or system operators. The local data centres serves as the primary hub for monitoring and managing data from all sensors, and receiving timely alerts enables it to take corrective measures quickly.

Furthermore, recognizing the potential severity of the situation, the system also sends an SOS alert to the nearest hospital. This dual alert mechanism—notification to both the local data centre and the nearest hospital—ensures a comprehensive and swift response to the detected anomaly. The alert to the hospital is particularly crucial in scenarios where sensor readings might indicate environmental hazards or other critical conditions that could impact public health. For instance, in a water distribution network, an anomaly could indicate contamination or a leak, necessitating immediate intervention to prevent health risks.

The system's ability to differentiate between transient anomalies and persistent issues ensures that only significant deviations trigger alerts, thereby avoiding false alarms. This robustness is essential for maintaining operational efficiency and ensuring that resources are directed towards genuine issues. By implementing this rigorous monitoring and alert system, the network ensures that any deviations from normal sensor readings are not only detected but also addressed swiftly and efficiently.

The combination of local data analysis and immediate SOS alerts provides a robust framework for maintaining operational safety and health standards, crucial for critical applications such as industrial monitoring, environmental safety, and healthcare management. By integrating a layered response strategy, the system ensures that not all anomalies trigger alarms, thereby avoiding unnecessary disruptions. The initial step involves filtering out transient anomalies, which helps in reducing false positives caused by temporary fluctuations or minor sensor errors. Subsequently, the system monitors these anomalies for persistence, ensuring that only consistent and significant deviations from the norm are flagged for further scrutiny.

If an anomaly persists and meets predefined thresholds, it escalates the issue, triggering immediate SOS alerts to notify relevant personnel. This escalation process is crucial as it guarantees that confirmed issues are communicated promptly to local authorities, healthcare providers, or any other designated responders, enabling them to take timely and appropriate actions. By doing so, the system effectively manages sensor data, ensuring operational safety and health standards are upheld without overwhelming stakeholders with false alarms.

This multi-tiered monitoring and alerting mechanism underscores the system's capability to handle critical situations with precision and efficiency. It ensures that the response to potential threats is proportional to their severity, thus optimizing resource allocation and response efforts. Overall, this well-thought-out approach not only enhances the reliability and efficiency of monitoring systems but also significantly contributes to the safety and well-being of the individuals and environments under its surveillance.

The SoS data is finally sent to nearest hospital using modified reverse Dijkstra algorithm can be effectively used to transmit data across a network by optimizing the path from multiple nodes to a single destination node, typically a data centre. Here's a step-by-step algorithm to achieve this:

First, initialize the network graph, where nodes represent sensors and edges represent communication links with associated transmission costs. Select the data centre as the source node. Initialize the distance to the source node as zero and all other nodes as infinity. Use a priority queue to manage the nodes to be processed, starting with the data centre.

While the priority queue is not empty, extract the node with the smallest distance (initially, the data centre). For each neighbouring node, calculate the potential new distance by adding the cost of the edge connecting them. If this new distance is smaller

than the currently known distance for the neighbouring node, update its distance and record the current node as its predecessor. Insert the neighbouring node into the priority queue with its updated distance.

Continue this process until all nodes have been processed, ensuring the shortest paths from all sensors to the data centre are found. Once the algorithm completes, each node will have a path leading to the data centre through the shortest possible route. Transmit data by following these recorded paths, ensuring minimal transmission cost and energy usage. This approach is particularly effective in sensor networks with limited energy resources, optimizing the data flow and prolonging the network's operational life.

To optimize data transmission from a any sensor, we are employing the concept of sending only deviations from a predefined normal range in our proposed algorithm can significantly reduce the number of bits required, thereby enhancing efficiency. Typically, a sensor would use 8 bits to represent heart rate values, which accommodates a range up to 255 beats per minute. However, this full range is excessive for most practical purposes, as normal human heart rates generally fall within 60 to 100 beats per minute. By focusing on deviations from this normal range, we can substantially reduce the amount of data transmitted.

For example, consider a heart rate of 115 beats per minute. Normally to send the actual data 115, we need 8 bits. Instead of transmitting the entire value, the sensor calculates the deviation from the upper limit of the normal range, which is 120. The deviation is $115 - 100 = 15$ beats per minute. This deviation can be represented using only 4 bits, as 4 bits can represent values from 0 to 15. Similarly, if the heart rate drops to 50 beats per minute, the deviation from the lower limit (60 beats per minute) is $60 - 50 = 10$ beats per minute, which also requires just 4 bits for representation. So, we will be getting only deviated value. But how we can know that the coming deviated data is more than 100 or less than 60? For that we are using the another 8 bit array (Deviated array). If respective sensor index has value 1, that suggest the deviated value is more than 100 and if the index is 0 then the opposite of that.

This approach drastically reduces the data payload. Instead of consistently using 8 bits to send every heart rate value, the system now often uses only 4 bits to convey the necessary information about deviations from the normal range. This reduction in data size translates to lower energy consumption for data transmission, which is particularly beneficial for battery-powered sensors that require long operational lifetimes. Moreover, it also conserves bandwidth, making the system more efficient in environments where multiple sensors transmit data simultaneously.

By implementing this deviation-based reporting strategy, the system ensures that it sends only essential information while maintaining the accuracy of heart rate monitoring. This method not only extends the battery life of the sensors but also enhances the overall efficiency of the monitoring system. It allows for continuous and reliable heart rate tracking with minimal data usage, which is crucial in applications such as remote patient monitoring, fitness tracking, and critical health surveillance. The practical implications of this approach are significant, providing a robust solution for efficient data management in sensor networks, ensuring that both data integrity and system longevity are upheld.

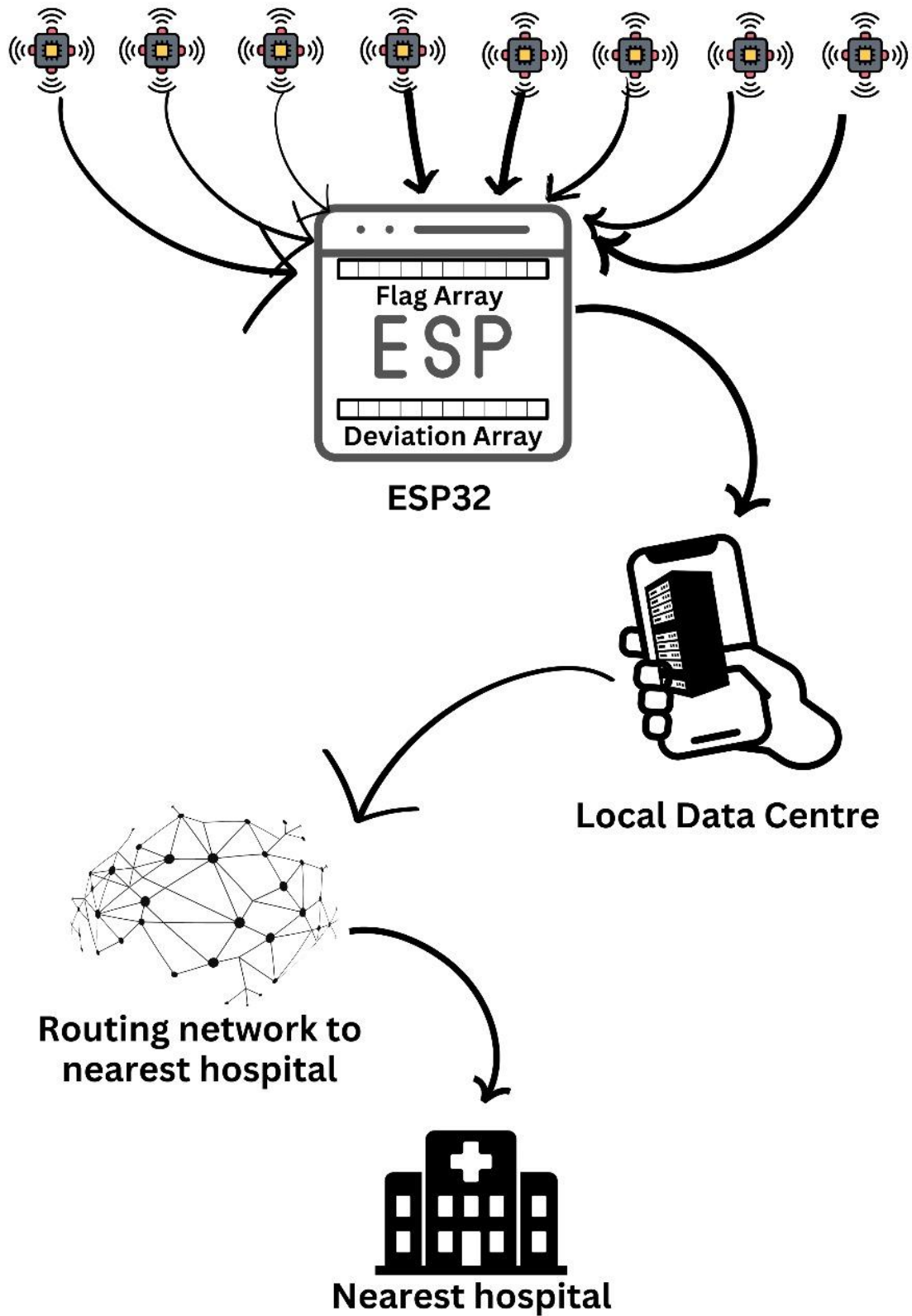


Figure 5: Data Flow and Alert Protocol in ESP32 Sensor Networks

CHAPTER 6

PERFORMANCE EVOLUATION

As we have already seen that according to our workflow our sensor data will be sent the packet to the local data centre when there's a abnormality in sensing values. So, performance of our algorithm depends on how much data we are sending through the path among data collector ESP32 and the local data centre.

To calculate total energy, we have to calculate the energy for data transmission from ESP32 and to receive the data at local data centre.

The energy required to transmit data, incorporating both the basic transmission energy and the additional energy due to the amplifier that compensates for the distance. Here's a breakdown of the energy calculation:

1. Energy required for transmission of data:

$$E_{tx} = E \times k + \varepsilon_{amp} \times k \times d^2$$

Where:

- E_{tx} = Total energy required for transmission
- E = Energy consumed per bit for basic transmission
- k = Number of bits to be transmitted
- ε_{amp} = Energy consumed per bit per square meter for the amplifier
- d = Distance between the transmitter and the receiver

Explanation:

1. Basic Transmission Energy ($E \times k$):

- This term represents the energy required to transmit k bits of data without considering the distance.
- E is the energy consumed per bit, and multiplying it by k gives the total energy for transmitting kk bits.

2. Amplifier Energy ($\varepsilon_{amp} \times k \times d^2$):

- This term accounts for the additional energy required due to the distance d between the transmitter and the receiver.
- ε_{amp} is the energy consumed per bit per square meter to account for the increased energy needed to transmit over a distance d .

- The term d^2 represents the squared distance, indicating that energy consumption increases with the square of the distance (a common model in wireless communication).

2. Energy required for receiving data:

$$E_{rx} = E \times k$$

Where:

- E_{rx} = Total energy required for receiving the data
- E = Energy consumed per bit by the receiver circuitry
- k = Number of bits to be received

3. Total Energy for entire process:

$$E_{total} = E_{tx} + E_{rx}$$

Explanation with Example:

Let's calculate for 1000 instances to generalize the calculation.

Let's first define all constants:

- $E=50$ nJ/bit
- $\varepsilon_{amp}=100$ nJ/bit/m²

Let, d be 1m for our calculation.

So, we will be having 2 cases of the entire scenario to compare the energy consumed by our algorithm to transmit and receive data and the energy consumed for regular data transmission and receive.

- **Case-1:** Energy consumed by regular method to transmit and receive
- **Case-2:** Energy consumed by our algorithm to transmit and receive data

Case-1:

k is volume of data and that depends on following factors without data optimization:

- We need to transfer 8bit GSM data to check still the values are abnormal (Flag array).

- For regular case, we need the send the real sensor data and for 2 more associated features along with the first abnormal feature we need 24 bits as all these three features may will have at most value 250 and that can be represented by 8 bits.

So, $k = (8+24) \text{ bits} = 32 \text{ bits}$.

Now, we plug these values into the formula for a instance:

1. Calculate Total Transmission Energy:

$$E \times k = 50 \text{ nJ/bit} \times 32 \text{ bits}$$

$$= 1600 \text{ nJ}$$

$$\varepsilon_{amp} \times k \times d^2 = 100 \text{ nJ/bit/m}^2 \times 32 \text{ bits} \times (1 \text{ meter})^2$$

$$= 3200 \text{ nJ}$$

$$E_{tx} = E \times k + \varepsilon_{amp} \times k \times d^2$$

$$= 1600 \text{ nJ} + 3200 \text{ nJ} = 4800 \text{ nJ}$$

2. Calculate Total Receiving Energy:

$$E_{rx} = E \times k$$

$$= 50 \text{ nJ/bit} \times 32 \text{ bits}$$

$$= 1600 \text{ nJ}$$

3, Total Energy:

$$E_{total} = E_{tx} + E_{rx}$$

$$= 4800 \text{ nJ} + 1600 \text{ nJ}$$

$$= 6400 \text{ nJ}$$

For 1000 instances, E_{total} becomes $1000 \times 6400 \text{ nJ} = 6400 \text{ } \mu\text{J}$.

Case-2:

k is volume of data and that depends on following factors in our algorithm with data optimization:

- We need to transfer 8bit GSM data to check still the values are abnormal (Flag array).
- Along with, we will send 8bit another array to check deviation in value towards upper side or lower side (Deviation array).

- From our experiment, we have got to send the deviation value of 2 more associated features along with the first abnormal feature, the deviated values need 7 bits for all three features.

So, $k = (8+8+7) \text{ bits} = 23 \text{ bits}$.

Now, we plug these values into the formula for a instance:

2. Calculate Total Transmission Energy:

$$E \times k = 50 \text{ nJ/bit} \times 23 \text{ bits}$$

$$= 1150 \text{ nJ}$$

$$\varepsilon_{amp} \times k \times d^2 = 100 \text{ nJ/bit/m}^2 \times 23 \text{ bits} \times (1 \text{ meter})^2$$

$$= 2300 \text{ nJ}$$

$$E_{tx} = E \times k + \varepsilon_{amp} \times k \times d^2$$

$$= 1150 \text{ nJ} + 2300 \text{ nJ} = 3450 \text{ nJ}$$

2. Calculate Total Receiving Energy:

$$E_{rx} = E \times k$$

$$= 50 \text{ nJ/bit} \times 23 \text{ bits}$$

$$= 1150 \text{ nJ}$$

3, Total Energy:

$$E_{total} = E_{tx} + E_{rx}$$

$$= 3450 \text{ nJ} + 1150 \text{ nJ}$$

$$= 4600 \text{ nJ}$$

For 1000 instances, E_{total} becomes $1000 \times 4600 \text{ nJ} = 4600 \mu\text{J}$.

Comparison of energy consumption by both methods:

Total energy using data optimization = $4600 \mu\text{J}$

Total energy without data optimization = $6400 \mu\text{J}$

So, reduction in energy consumption by our proposed algorithm

$$= \frac{(\text{Total energy without data optimization}) - (\text{Total energy using data optimization})}{\text{Total energy without data optimization}} \times 100\%$$

$$= \frac{6400 - 4600}{6400} \times 100\%$$

$$= 28.125\%$$

So, our proposed algorithm consumes 28.125% less energy in this case.

So, basically our algorithm works perfectly with higher number of sensors.

We have performed the comparative energy calculation for several times with values obtained from our experiment and have put them into a graph. Here's the graph:

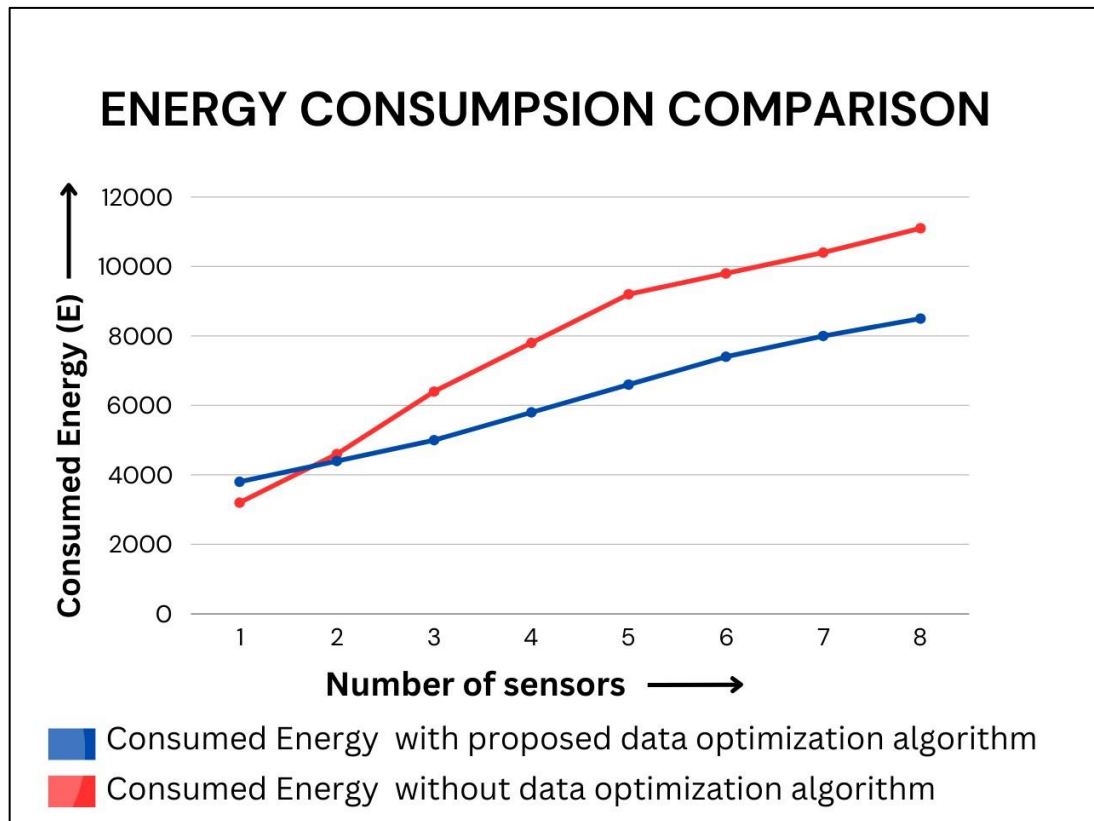


Figure 6: Energy comparison graph

CHAPTER 6

FUTURE WORK

Compressed sensing, a revolutionary signal processing technique, has garnered significant attention in recent years due to its ability to efficiently acquire and reconstruct signals from a limited number of measurements. This innovative approach has found applications in various fields such as medical imaging, communication systems, and remote sensing, offering a promising solution to the challenges of traditional signal processing methods. As we delve into the future of compressed sensing, it is essential to explore the current trends, challenges, and potential advancements that will shape its trajectory in the coming years.

Compressed sensing has witnessed rapid growth and adoption across diverse domains, driven by its capability to reduce data acquisition and processing requirements significantly. In medical imaging, for instance, compressed sensing has enabled faster MRI scans with reduced scan times, minimizing patient discomfort and enhancing imaging quality. Similarly, in wireless communication systems, compressed sensing has facilitated efficient data transmission and improved spectrum utilization by compressing signals before transmission and reconstructing them at the receiver end.

Moreover, the integration of compressed sensing with machine learning algorithms has opened up new avenues for signal reconstruction and processing. By leveraging the power of deep learning and neural networks, researchers have achieved remarkable results in reconstructing high-quality images and signals from sparse measurements, pushing the boundaries of what is achievable with traditional methods.

Despite its promising potential, compressed sensing faces several challenges that need to be addressed to realize its full capabilities. One of the primary challenges is the computational complexity involved in signal reconstruction, especially for high-dimensional data. As the size and complexity of datasets increase, the computational

burden of solving optimization problems in compressed sensing also escalates, necessitating efficient algorithms and hardware implementations.

Another challenge lies in the design of optimal sensing matrices that enable accurate signal recovery. The choice of sensing matrix plays a crucial role in the success of compressed sensing algorithms, and developing robust and adaptive sensing strategies remains a key research area. Additionally, ensuring the robustness of compressed sensing techniques to noise, artifacts, and model uncertainties is essential for real-world applications where data may be corrupted or incomplete.

Despite these challenges, the future of compressed sensing holds immense opportunities for innovation and advancement. Emerging technologies such as quantum computing and sparse signal processing offer new avenues for enhancing the efficiency and scalability of compressed sensing algorithms. By harnessing the power of quantum computing to solve optimization problems more efficiently, researchers can potentially accelerate the reconstruction process and handle larger datasets with ease.

Looking ahead, several key areas are poised to shape the future of compressed sensing and drive its evolution towards more robust and versatile applications. One such area is the development of adaptive sensing schemes that can dynamically adjust the sensing process based on the characteristics of the input signal. By incorporating feedback mechanisms and adaptive algorithms, researchers aim to improve the efficiency and accuracy of signal recovery in dynamic and changing environments.

Furthermore, the fusion of compressed sensing with other emerging technologies such as Internet of Things (IoT) and edge computing holds promise for enabling real-time signal processing and analysis in resource-constrained environments. By deploying compressed sensing algorithms on edge devices and IoT sensors, data can be

processed locally, reducing latency and bandwidth requirements while preserving privacy and security.

Moreover, the integration of compressed sensing with advanced imaging modalities such as hyperspectral imaging and multi-modal imaging presents exciting opportunities for enhancing diagnostic capabilities in healthcare and remote sensing applications. By combining the strengths of compressed sensing with advanced imaging techniques, researchers can achieve higher spatial and spectral resolution, enabling more accurate and detailed image reconstruction.

In conclusion, the future of compressed sensing is bright, with ongoing research efforts focused on addressing current challenges and exploring new frontiers in signal processing and data acquisition. By leveraging advancements in machine learning, quantum computing, and adaptive sensing strategies, compressed sensing is poised to revolutionize how signals are acquired, processed, and reconstructed in diverse applications. As researchers continue to push the boundaries of innovation, the potential for compressed sensing to transform industries and enable new capabilities remains a compelling prospect in the realm of signal processing and data science.

CHAPTER 7

CONCLUSION

In conclusion, our research delved into the pressing issue of energy consumption in wireless sensor networks, recognizing the inherent challenges posed by the limited resources of sensor nodes. By focusing on innovative data compression techniques, we aimed to tackle these challenges head-on and extend the overall lifespan of these networks.

The strategic reduction of data size before transmission emerged as a key solution, alleviating the strain on resources and enhancing the efficiency of wireless sensor networks. Our findings underscore the significance of data compression in addressing the complexities associated with these networks, particularly in the context of the growing Internet of Things (IoT) landscape.

Looking ahead, future possibilities abound in further refining and implementing these data compression techniques. Continued research and development could lead to even more efficient ways of reducing energy consumption, making wireless sensor networks more sustainable and resilient. Exploring applications beyond the scope of our current study could unveil new opportunities, potentially impacting various sectors and contributing to the advancement of IoT technologies.

We express our gratitude to our mentors for their guidance and support throughout this project. Their insights have been invaluable in shaping our research and driving us towards meaningful solutions. As we conclude this endeavor, we look forward to the ongoing efforts in this field, optimistic about the positive impact that innovations in data compression can bring to the realm of wireless sensor networks and the broader landscape of the Internet of Things.

CHAPTER 8

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